

MedGuard AI – Member Task Distribution Document

● Phase 1 – Requirement & Planning (All Members – 1 Week)

- **Backend Lead:** Finalize API structure, define endpoint contracts, decide authentication flow (JWT).
- **AI/ML Lead:** Finalize embedding model, decide FAISS index type, research LLM model for Ollama.
- **DB & DevOps Lead:** Design initial DB schema, define ENUM types, plan Docker container structure.
- **Frontend Lead:** Create UI wireframes, define user flow (verification + symptom module).

Deliverable: Architecture Diagram + ER Diagram + Flow Design

● Phase 2 – Database & Dataset Setup

- **DB & DevOps Lead (Primary Owner):** Create PostgreSQL schema (Users, Medicines, Barcodes, Embedding Metadata, Verification History), add indexes, implement models & migrations.
- **AI/ML Lead:** Clean medicine dataset, assign legality status, prepare text chunks for embedding.
- **Backend Lead:** Connect Django to PostgreSQL, create models & migrations.
- **Frontend Lead:** Start basic authentication UI (Login/Register).

Deliverable: Working DB + Clean Dataset

● Phase 3 – RAG & FAISS Setup (AI Core Phase)

- **AI/ML Lead (Primary Owner):** Generate embeddings, build FAISS index, implement similarity search, tune threshold, test <5 sec search target.
- **DB & DevOps Lead:** Setup FAISS persistent Docker volume, store embedding metadata mapping.
- **Backend Lead:** Create RAG search API endpoint, integrate FAISS service.
- **Frontend Lead:** Create medicine text input UI, display similarity score.

Deliverable: Working RAG Search API

● Phase 4 – Medicine Verification Service

- **Backend Lead (Primary Owner):** OCR integration, barcode decoding, expiry extraction logic, authenticity classification logic, result ENUM handling.
- **AI/ML Lead:** Validate similarity scoring, assist in decision logic tuning.
- **DB & DevOps Lead:** Implement verification history logging, optimize indexing.
- **Frontend Lead:** Multi-modal upload UI, display color-coded results (Authentic, Fake, Expired, Illegal).

Deliverable: Full Medicine Verification Module

● Phase 5 – Symptom Remedy LLM Module

- **AI/ML Lead (Primary Owner):** Ollama setup, model selection, strict prompt design, enforce disclaimer.
- **Backend Lead:** Create symptom API endpoint, handle timeout & error control.
- **DB & DevOps Lead:** Optional logging table, resource optimization (RAM monitoring).
- **Frontend Lead:** Symptom input UI, display Remedy, Proportions, Precautions, Disclaimer.

Deliverable: Working LLM Module

● Phase 6 – Dockerization & Deployment

- **DB & DevOps Lead (Primary Owner):** Write Dockerfiles, Setup Docker Compose (api-gateway, medicine-service, symptom-llm-service, postgres, ollama), configure persistent volumes, Cloud deployment (AWS/DigitalOcean).
- **Backend Lead:** Environment configs, Production settings, Secure JWT handling.
- **AI/ML Lead:** Optimize embedding memory usage, tune FAISS index.
- **Frontend Lead:** Production build, connect to deployed APIs.

Deliverable: Fully Dockerized Cloud-Ready System

 Phase 7 – Testing & Optimization

- **Backend Lead:** API testing, error handling tests, security validation.
- **AI/ML Lead:** Similarity threshold tuning, LLM hallucination checks, performance measurement.
- **DB & DevOps Lead:** Load testing (100 users), backup setup, monitoring setup.
- **Frontend Lead:** UX testing, responsiveness testing, Android compatibility testing.

Deliverable: Stable Production Version

Clear Responsibility Matrix (Key Modules)

Module	Primary Owner	Support
API Gateway	Backend ▾	DevOps ▾
PostgreSQL Design	DB Lead ▾	Backend ▾
RAG + FAISS	AI Lead ▾	Backend ▾
OCR + Barcode	Backend ▾	AI ▾
Authenticity Logic	Backend ▾	AI ▾
Symptom LLM	AI Lead ▾	Backend ▾
Docker Deployment	DevOps ▾	Backend ▾
UI/UX	Frontend ▾	Backend ▾

MedGuard AI – 4 Member Task Distribution (Role Summary)

Person	Role	Main Responsibility
Person 1	Backend	APIs + Medicine Logic (API Gateway, Verification Logic, Symptom API, Docs)
Person 2	AI/ML	RAG + FAISS + LLM (Dataset, Embedding, FAISS Setup, Ollama Integration)
Person 3	DB & DevOps	PostgreSQL + Docker + Deployment (DB Tables, FAISS)

		Persistence, Dockerization, Cloud Setup)
Person 4	Frontend	UI + Integration (UI Design, Multi-Modal Input, Result Display, Responsive Design)